MX v1.1: Advancing Open Standards for Low-Precision AI with MicroXcaling formats

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Collaborators

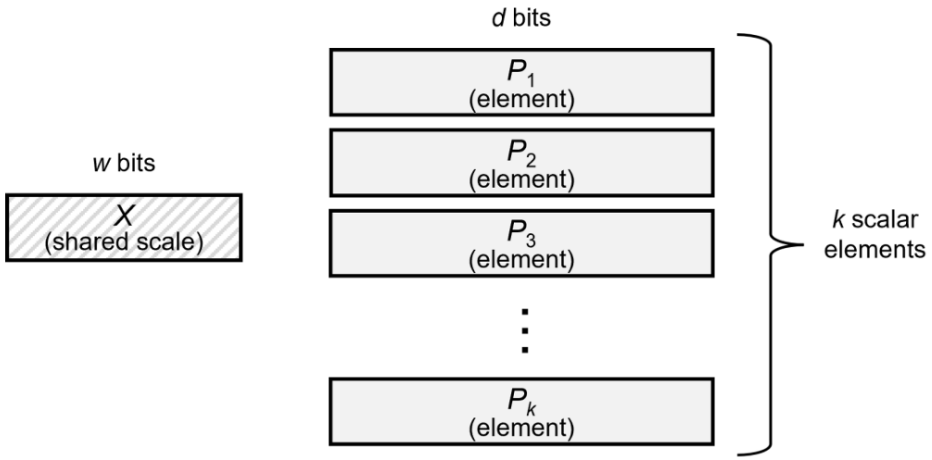
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Executive Summary

- **OCP MX: Open, interoperable AI numeric formats**
  - OCP MicroXcaling (MX) defines an open, interoperable family of low-precision data formats for efficient AI training and inference.
- **Built through industry collaboration**
  - MX is developed collaboratively by leading AI hardware and software companies to reduce fragmentation and improve portability.
- **Industry demand is accelerating**
  - Since MX 1.0, adoption of low-precision AI computation has accelerated, increasing demand for common formats and implementation guidance.
- **MX 1.1 strengthens the specification**
  - MX 1.1 improves clarity, refines definitions, and provides implementation guidance while preserving interoperability.
- **Enabling a portable AI ecosystem**
  - Standardized low-precision formats and operations enable a portable, interoperable, and scalable AI ecosystem.

OCP MX 1.0 Spec

- An MX-compliant format is characterized by three components
  - Shared scale (X)
  - Private elements (Pi)
  - Scaling block size (k)



Format Name	Element Data Type	Element Bits (d)	Scaling Block Size (k)	Scale Data Type	Scale Bits (w)
MXFP8	FP8 (E5M2)	8	32	E8M0	8
	FP8 (E4M3)				
MXFP6	FP6 (E3M2)	6	32	E8M0	8
	FP6 (E2M3)				
MXFP4	FP4 (E2M1)	4	32	E8M0	8
MXINT8	INT8	8	32	E8M0	8

OCP MX 1.1 Spec – new updates!

1. Description for conversion from vector of scalar elements to MX-compliant format in Section 6.3 updated.
2. Renamed E8M0 to UE8M0.
3. Refinements to FP8, FP6, and FP4 conversion modes and rounding terminology in Section 5.3.
4. Clarifications of edge-cases for NaN (Section 5.1) and out-of-range (Sections 5.1, 6.1, 6.2) behavior.
5. Various minor grammatical and wording changes.

Update 1: Conversion to MX-compliant Format

- Introduced $\mathrm{expAdj(V)}$, an additional term in $\mathrm{SF(V)}$, scale factor calculation
- The added $\mathrm{expAdj(V)}$ term may be used to reduce errors introduced whenever $\mathrm{max_{V_i \in V}(|Vi|)}$ saturates after scaling and quantizing to the element data type.
- The functional details of $\mathrm{expAdj(V)}$ are implementation-defined and may optimize rounding or quantization noise across the converted block

$$\mathrm{SF(V)} = 2^{\left\lfloor \log_2\left(\max_{V_i \in V}(|V_i|)\right) - maxExp + expAdj(V) \right\rfloor}$$

Update 2: Renamed E8M0 to UE8M0

- UE8M0 is an unsigned FP8 representation closely related to a conventional biased Float32 exponent.
- Unlike Float32 exponents, there is no representation for Inf or Zero and only a single NaN encoding is reserved.
- UE8M0 provides an additional step of dynamic range when compared to a Float32 exponent.

UE8M0	
Exponent Bias	127
Supported Exponent Range	-127 to 127
Infinities	N/A
NaN	11111111_2
Zeros	N/A

Update 3 and 4: Refinements to FP8, FP6, and FP4 conversion modes

- Better alignment with OCP 8-bit floating point (OFP8) specification
- During FP8 conversion, implementations must support the saturate (SAT) and non-saturate (NONSAT) overflow modes defined in the table below.
- For FP6 and FP4, conversion semantics are simplified (see spec for details).

Source Value	Destination Value			
	E4M3		E5M2	
	SAT	NONSAT	SAT	NONSAT
NaN	NaN	NaN	NaN	NaN
±Inf	±max_E4M3	NaN	±max_E5M2	±Inf
Greater than the max destination OFP8 magnitude when rounded to the destination precision	±max_E4M3	NaN	±max_E5M2	±Inf

Evolving OCP MX for Next-Generation AI

- **MX will continue to evolve**
  Future releases will adapt to emerging AI workloads and industry requirements.
- **End-to-end alignment is essential**
  AI formats must be consistent across hardware, compilers, frameworks, and libraries.
- **Standards must be defined early**
  Long software lead times require stable format specifications.
- **Proprietary formats create fragmentation**
  Multiple vendor-specific formats increase duplication and market confusion.
- **Developers need clear specifications**
  Well-defined interfaces simplify adoption and interoperability.
- **Open standards enable innovation**
  OCP MX provides a portable, interoperable foundation for the AI ecosystem.

Industry Adoption Across Hardware and Software

MX format Software Enablement
PyTorch – ML Framework
MX PyTorch Emulation Library
cuDNN – Deep Neural Network Library
CUTLASS – Templates for Linear Algebra
OpenVINO – Toolkit for AI Inference
Triton - Kernels
ONNX - ecosystem

MX format Hardware Enablement
Microsoft Maia Series
AMD MI300 Accelerator Series
x86 AI Compute Extensions (ACE) v1
Intel Gaudi 2 and 3 AI accelerators
Nvidia Blackwell GPU
Qualcomm Cloud AI 100 Accelerator
AWS Trainium3
d-Matrix Corsair

Benefits of OCP MX formats

- Low-precision formats deliver near-lossless accuracy with significantly fewer bits
  - Improve performance, silicon efficiency, and energy efficiency
  - Increase effective memory bandwidth and storage capacity
- Compute cost scales quadratically with precision, making lower-bit formats highly efficient
- Memory bandwidth and capacity scale linearly with bit-width
- Enable continued AI scaling despite slowing transistor advancements
- Provide a critical path to sustaining accelerator performance and efficiency gains

Call to Action

- The OCP MX specifications are available at the OCP contribution portal
  - https://www.opencompute.org/documents/OCP_MicroXcaling_Spec_v1.1.pdf
- Use these standard formats in your AI applications and hardware implementations
  - MX emulator v1.1.0 https://github.com/microsoft/microxcaling
- MX is continuing to evolve, for inquiries contact:
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Thank You!

